EXPERIMENT – 8

Privilege Escalation Using TryHackMe Platform

Aim

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)

Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.
In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.
TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:
2. whoami
3. Check system information:

4. uname -a
5. Check sudo permissions:
6. sudo -l

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

find / -perm -4000 2>/dev/null

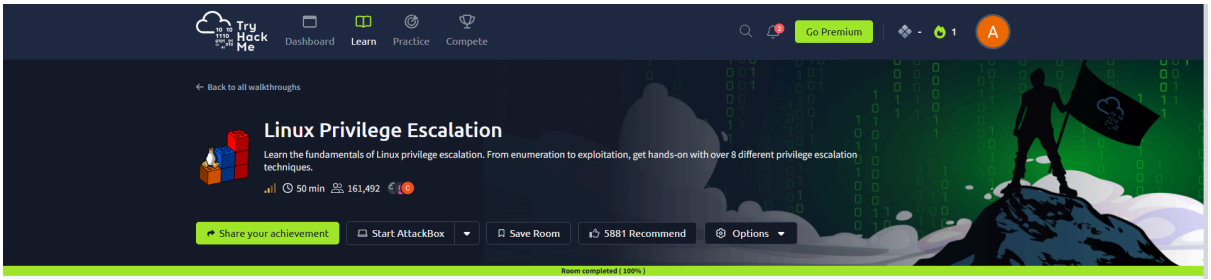
Step 6: Exploit Vulnerability

1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output




```
Last login: Wed Apr 15 15:31:08 2026 from ip-10-48-126-153.ap-south-1.compute.internal
[leonard@ip-10-48-136-143 ~]$ whoami
leonard
[leonard@ip-10-48-136-143 ~]$ uname -a
Linux ip-10-48-136-143 3.10.0-1160.el7.x86_64 #1 SMP Mon Oct 19 16:18:59 UTC 2020 x86_64 x86_
64 x86_64 GNU/Linux
[leonard@ip-10-48-136-143 ~]$
```

```
[leonard@ip-10-49-148-234 ~]$ find / -perm -4000 2>/dev/null
/usr/bin/base64
/usr/bin/ksu
/usr/bin/fusermount
/usr/bin/passwd
/usr/bin/gpasswd
/usr/bin/chage
/usr/bin/newgrp
/usr/bin/staprun
/usr/bin/chfn
/usr/bin/su
/usr/bin/chsh
/usr/bin/Xorg
/usr/bin/mount
/usr/bin/umount
/usr/bin/crontab
/usr/bin/pkexec
/usr/bin/at
/usr/bin/sudo
/usr/sbin/pam_timestamp_check
/usr/sbin/unix_chkpwd
/usr/sbin/usernetctl
/usr/sbin/userhelper
/usr/sbin/mount.nfs
/usr/lib/polkit-1/polkit-agent-helper-1
```

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform.
The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.